# Section 7 solution

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## 3. Exercises

### 3.1 Expenditure Minimization

Consider a consumer with Cobb-Douglas preferences:

$$u(x_1, x_2) = \alpha \ln(x_1) + (1-\alpha)\ln(x_2)$$

1. Derive the Hicksian demand functions: $h_i^*(p_1, p_2, U_0)$.
2. Derive the expenditure function: $e(p_1, p_2, U_0)$.
3. Derive the Marshallian demand functions: $x_i^*(p_1, p_2, M)$.
4. (Optional) Derive the indirect utility function: $v(p_1, p_2, M)$
5. Verify that $x_i^*(p_1, p_2, e(p_1, p_2, U_0)) = h_i^*(p_1, p_2, U_0)$.

**Answer: We want to solve:**

$$\min_{h_1, h_2} p_1 h_1 + p_2 h_2 \quad s.t.\, u(h_1, h_2) = \alpha \ln h_1 + (1-\alpha)\ln h_2 = \bar{u}$$

**So your FOCs are:**

$$p_1 + \lambda \frac{\alpha}{h_1} = 0$$
$$p_2 + \lambda \frac{1-\alpha}{h_2} = 0$$
$$\alpha \ln h_1 + (1-\alpha)\ln h_2 - \bar{u} = 0$$

**Combining the first two FOCs yields:**

$$\frac{p_1 h_1}{\alpha} = \frac{p_2 h_2}{1-\alpha}$$

**Substituting into the last FOC yields:**

$$\alpha\left[\ln\left(\frac{p_2}{p_1}\frac{\alpha}{1-\alpha}h_2\right)\right] + (1-\alpha)\ln h_2 - \bar{u} = 0$$
$$\alpha\left[\ln\left(\frac{p_2}{p_1}\right) + \ln\left(\frac{\alpha}{1-\alpha}\right) + \ln h_2\right] + (1-\alpha)\ln h_2 - \bar{u} = 0$$

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**Which becomes:**

$$\ln h_2 = \bar{u} + \alpha \ln \frac{p_1}{p_2} + \alpha \ln \frac{1-\alpha}{\alpha}$$
$$h_2^* = \left(\frac{p_1}{p_2}\frac{1-\alpha}{\alpha}\right)^{\alpha} e^{\bar{u}}$$

**Plug this back to FOC and solve for $h_1^*$:**

$$h_1^* = \left(\frac{p_2}{p_1}\frac{\alpha}{1-\alpha}\right)^{1-\alpha} e^{\bar{u}}$$

**Our expenditure function is thus:**

$$\begin{aligned} e\left(p_1, p_2, \bar{u}\right) &= p_1 \left(\frac{p_2}{p_1}\frac{\alpha}{1-\alpha}\right)^{1-\alpha} e^{\bar{u}} + p_2 \left(\frac{p_1}{p_2}\frac{1-\alpha}{\alpha}\right)^{\alpha} e^{\bar{u}} \\ &= e^{\bar{u}} \left[p_1^{\alpha}\left(p_2\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + p_2^{1-\alpha}\left(p_1\frac{1-\alpha}{\alpha}\right)^{\alpha}\right] \\ &= e^{\bar{u}} p_1^{\alpha} p_2^{1-\alpha}\left[\left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha}\right)^{\alpha}\right] \end{aligned}$$

**Recall that the Marshallian demand functions for Cobb-Douglas are:**

$$\begin{aligned} x_1^* &= \frac{\alpha}{p_1} M \\ x_2^* &= \frac{1-\alpha}{p_2} M \\ v(p_1, p_2, M) = U(x_1^*, x_2^*) &= \alpha \log(\frac{\alpha}{p_1} M) + (1-\alpha)\log(\frac{1-\alpha}{p_2} M) \end{aligned}$$

**Verifying the dual condition means plugging the expenditure function into the Marshallian demand functions and verifying that they equal the Hicksian demand functions. Let's show this for $x_1$:**

$$\begin{aligned} x_1^* &= \frac{\alpha}{p_1}\left\{e^{\bar{u}} p_1^{\alpha} p_2^{1-\alpha}\left[\left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha}\right)^{\alpha}\right]\right\} \\ &= e^{\bar{u}}\left(\frac{p_2}{p_1}\right)^{1-\alpha}\left[\frac{\alpha^{2-\alpha}}{(1-\alpha)^{1-\alpha}} + \frac{(1-\alpha)^{\alpha}}{\alpha^{\alpha-1}}\right] \\ &= e^{\bar{u}}\left(\frac{p_2}{p_1}\right)^{1-\alpha}\left[\frac{\alpha + (1-\alpha)}{(1-\alpha)^{1-\alpha}\alpha^{\alpha-1}}\right] \\ &= e^{\bar{u}}\left(\frac{p_2}{p_1}\right)^{1-\alpha}\left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} \\ &= \left(\frac{p_2}{p_1}\frac{\alpha}{1-\alpha}\right)^{1-\alpha} e^{\bar{u}} \\ &= h_1^* \end{aligned}$$

**Not surprising.**

## 3.2 Practice: Utility Maximization & Indirect Utility Functions (Midterm Fall 2005)

Consider Mario, a Californian resident with income $M$. Mario cares about own consumption $c$ and about a charitable project $G$ in Louisiana. The welfare effects of the project for Louisana citizens are measured by $h(g+S)$, where $g$ is Mario's charitable donation and $S$ is the seed money donated by others. Mario derives utility $\alpha h(g+S)$ from the project, with $\alpha \geq 0$. Therefore Mario maximizes

$$\begin{aligned}
&\max_{c,g} u(c) + \alpha h(g+S) \\
&s.t.\ c + g \leq M, c \geq 0, g \geq 0
\end{aligned}$$

We assume $u'(\cdot) > 0$, $u''(\cdot) < 0$, $h'(\cdot) > 0$, and $h''(\cdot) < 0$. That is, both $u$ and $h$ are increasing, concave functions. The function $h$ is concave because there are diminishing returns to the charitable project. We also assume $\alpha, S, M \geq 0$.

1. What is the interpretation of $\alpha$ in the utility function? Interpret in particular the special case $\alpha = 0$. **Answer: $\alpha$ weights how much Mario cares about the utility of Louisianans. If $\alpha = 0$, then Mario only cares about his own utility, not at all those of the Louisianans.**

2. Argue that the problem can be rewritten as

$$\begin{aligned}
&\max_{g} u(M-g) + \alpha h(g+S) \\
&s.t.\ 0 \leq g \leq M
\end{aligned}$$

**Answer: Mario's utility is monotonic based on our assumptions about the functional forms of Mario's subutilities $u'(\cdot)$ and $h'(\cdot)$ and how they relate to each other (i.e. additively). Therefore, Mario will spend all his money, which means he's not really making two choices, he's making only the choice of what portion to consume and what portion to give away.**

3. Neglecting the constraint $0 \leq g \leq M$ for now, derive the first- and second-order conditions. Argue that the point identified by the first-order conditions is a maximum.
**Answer: Our first order conditions are given by an unconstrained maximization problem:**

$$0 = -u'(M-g^*) + \alpha h'(g^*+S)$$

**and our second order conditions are**

$$0 > u''(M-g^*) + \alpha h''(g^*+S)$$

**We know that $u'' < 0$ and $h'' < 0$, so our second-order conditions are satisifed since $(M-g^*)$ and $(g^*+S)$ are positive.**

4. Solve for $g^*$ and $c^*$ using the first-order conditions for the case $u(c) = \log(c)$ and $h(g) = \log(g)$.
**Answer: Our first-order conditions become**

$$\begin{aligned}
0 &= -\frac{1}{M-g^*} + \frac{\alpha}{g^*+S} \\
\alpha M - \alpha g^* &= g^* + S \\
g^* &= \frac{\alpha M - S}{1+\alpha}
\end{aligned}$$

**It follows immediately that**

$$c^* = M - g^* = M - \left(\frac{\alpha M - S}{1+\alpha}\right) = \frac{M+S}{1+\alpha}$$

5. Keep assuming $u(c) = \log(c)$ an $h(g) = \log(g)$. Now it is time to check the constraint $0 \leq g \leq M$. Is the constraint $g \leq M$ always satisfied? Is the constraint $g \geq 0$ always satisfied? If not, provide an example to the converse. Write the solution for $g^*$ and $c^*$, taking into account corner solutions.
**Answer: First let's check the upper constraint. $g^*$ is decreasing in $S$, so let's check the smallest possible $S = 0$. In this case $g^* = \frac{\alpha M}{1+\alpha}$. Since $\frac{\alpha}{1+\alpha} \leq 1$, $g^* \leq M$.**

**Next, let's check the non-negativity constraint. Here, we can clearly see if $S$ is big enough (particularly if $S > \alpha M$), then Mario will want to give a negative amount. However, because only the non-negativity constraint can ever be binding, we only have a single corner solution in the event that $S > \alpha M$.[1] This gives a general solution of**

$$g^* = \max\left(\frac{\alpha M - S}{1+\alpha}, 0\right)$$
$$c^* = \min\left(\frac{M + S}{1+\alpha}, M\right)$$

6. Keep assuming $u(c) = \log(c)$ and $h(g) = \log(g)$. How do $c^*$ and $g^*$ depend on $M$, $\alpha$, and $S$? Provide intuition for each of these comparative statics, in particular for the latter.
   **Answer: From (4) we can see that increases in altruism $\alpha$ increase $g^*$ and decrease $c^*$, except if the solution is a corner solution. Not surprisingly, more altruisitic people prefer to spend more money on charity. Increases in income $M$ lead to an increase in both the consumption $c^*$ and the charitable contribution $g^*$. Not surprisingly, more income leads to more spending. Finally, a higher seed donation $S$ leads to lower charitable contribution and to more consumption. If other people have already contributed to the charity, an individual prefers to spend the money on own consumption.**

7. Now we go back to the general formulation with $u(\cdot)$ and $h(\cdot)$ concave. Using the implicit function theorem, and neglecting corner solutions, show that $\partial g^*/\partial S < 0$. Comment on this result.
   **Answer: Using the first-order conditions, we can use the IFT for univariate functions:**

$$\frac{\partial g^*}{\partial S} = -\frac{\frac{\partial f}{\partial S}}{\frac{\partial f}{\partial g}}$$
$$= -\frac{\alpha h''(g^* + S)}{u''(M - g^*) + \alpha h''(g^* + S)}$$

   **which is negative since both the numerator and denominator are negative (concavity of $u(\cdot)$ and $h(\cdot)$). It follows that $\frac{\partial g^*}{\partial S} < 0$ holds generally, not just for the case above.**

8. (Open-ended) An economist, John List, does an empirical study of the effect of seed money (the money collected early in a fundraising drive) on charitable donations. He finds that higher seed money is associated with *higher* donations, against the prediction in point 7. How can we explain this finding? **Idea: A likely explanation for this finding (see List and Lucking-Reiley, *Journal of Political Economy*, 2001) is that individuals use the seed money as a signal for the value of the charity. If others donated, it must be a worthwhjile charity. That is, high $S$ singals that the function $h(g)$ is larger than the donar previously thought. To the extent that this force is stronger than the one in point &, we can explain a positive correlation.**

9. Keep assuming $u(\cdot)$ and $h(\cdot)$ concave. Consider the indirect utility $V(S, M, \alpha) = u(M - g^*(S, M, \alpha)) + \alpha h(g^*(S, M, \alpha) + S)$. Use the envelope theorem to compute $dV(S, M, \alpha)/dS$, that is, how the indirect utility vary as the seed money increases. Under what conditions is $\partial V(S, M, \alpha)/\partial S > 0$ and why?
   **Using the envelope theorem, we know that**

$$\frac{dV}{dS} = \frac{\partial(u(\cdot) + \alpha h(\cdot))}{\partial S}$$

   **where we can neglect the indirect effects of S on $g^*$. It follows that**

$$\frac{dV}{dS} = \alpha h'(g^*(S, M, \alpha) + S)$$

[1] Recall that the Kuhn-Tucker complimentary slackness condition requires at least one of the non-negativity constraints to be binding.

**which is positive as long as $\alpha$ is positive. Therefore, any altruisitic individual is made better off when others contribute seed money.**